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Intervals, Adaptations, Metabolic, Psychological, Perceptual, Motivation

A full-page photograph of a woman running up a set of wide wooden stairs. She is captured from a low-angle, rear perspective, emphasizing her upward motion. She is wearing a white tank top, black leggings, and colorful sneakers. Her long hair is caught in a strong wind, blowing upwards and backwards in a dynamic arc. The scene is bathed in the warm, golden light of a low sun, likely during sunrise or sunset, which creates a soft, hazy atmosphere. The sky is a pale, warm tone, and the wooden planks of the stairs lead the eye diagonally across the frame.

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SIDEBAR: INTERVAL TERMINOLOGY

Interval Training: exercise that involves periods of intense work and recovery that vary on the intensity and duration of each segment.

Body Weight or Resistance HIIT: interval training that uses bodily movements, weighted objects, bars, or devices for high-repetition resistance activities.

Aerobic HIIT: interval training that uses traditional aerobic exercise modalities such as running and cycling.

programs like Tabata, CrossFit, boot camp training, or other similar classes. Although both types of HIIT programs are used widely and research indicates the effectiveness of body weight-type HIIT programming (13), most research has focused on HIIT that primarily is aerobic because cycling and treadmill running enable more accurate assessment of work to describe the training stimulus. Therefore, this article provides an overview of common protocols and evidence-based strategies to support designing effective programming using aerobic HIIT.

CONTEMPORARY RESEARCH MODELS OF INTERVAL TRAINING

Whereas modern interval training has linkages to military preparedness and athletic competition, the recent resurgence of interest in interval training can be attributed to potential health-related benefits of HIIT. Although a variety of aerobic HIIT training methods have emerged, research related to aerobic HIIT is linked primarily to three different models that vary on intensity, duration, and total exercise volume.

Wingate Model

Much of the current attention directed towards HIIT is linked to an approach that incorporates several trials of the famed Wingate Anaerobic Test. A single Wingate trial involves all-out, supramaximal cycling for a period of 30 seconds and remains the gold standard for assessing anaerobic capacity. Use of Wingate-based protocols within HIIT research requires the exerciser to complete several sprints separated by 4 minutes of unloaded pedaling (4). This HIIT training model typically is referred to as “low volume” because the total amount of work and time spent exercising is low (see Sidebar). Importantly, this approach to HIIT primarily was developed to demonstrate the potency of interval training in producing rapid cardiometabolic adaptations and is not generally the style of training recommended for a long-term program.

SIDEBAR: SPEED VS. GRADE

Each 1 mph increase in speed is roughly equal to 4% grade increase. This means that exercisers can pick and choose whether to increase work by way of speed or climbing a hill.

Scandinavian Model

Another prominent style of HIIT developed for cardiac patients incorporates several 4-minute intervals separated by an easy recovery of similar duration (18). As such, this training model is considered “high-volume” interval training because total time spent doing heavy exercise typically exceeds 15 minutes and total time is similar to traditional endurance training approaches. Importantly, intervals are performed somewhat below the maximal heart rate and are therefore not “all-out” exercise. Results from these studies generally demonstrate superior cardiovascular benefits of HIIT compared with standard aerobic exercise training (18).

Practical Model

A more recent variation of HIIT was developed as an alternative to the all-out intensities linked to the Wingate Model that maintains time efficiency. This approach involves performing 10 intervals on a cycle ergometer at intensities near peak work capacity for 60 seconds alternated with easy recovery intervals of the same duration (10). Given that this approach to interval training represents a midpoint of sorts between the two models above with respect to intensity, recovery, and total volume, this practical model can be considered a “medium-volume” interval training program. Whereas the acute and chronic tolerability of any interval-based training program requires additional research, a training approach based around a more practical model is appropriate for much of the general population, including type 2 diabetes mellitus (DM) patients (10).

PHYSICAL ACTIVITY GUIDELINES AND INTERVAL TRAINING

An important issue related to HIIT is the consideration of how such training can be incorporated within a PA program designed to meet current public health recommendations. Interval training does not explicitly “fit” within PA guidelines in a traditional sense. Although the work interval portion exceeds the intensity threshold for vigorous exercise, most aerobic HIIT sessions include no more than 10 to 12 minutes at this intensity. Similarly, most interval training sessions use relatively easy recovery segments that are best described as low intensity. As such, a typical interval training session does contribute toward meeting PA recommendations but primarily via vigorous intensities. Meeting the goal of 75 minutes of vigorous activity per week may prove challenging through HIIT alone. An important, yet presently unanswered, question is how to best substitute HIIT for more traditional forms of exercise for promoting health. Although the answer to this question is not yet fully settled, it is becoming increasingly clear that aerobic HIIT provides numerous health and fitness benefits.

PHYSIOLOGICAL BENEFITS

Research investigating aerobic HIIT-styled training indicates improvements in markers of cardiovascular health, metabolic capacity, and aerobic performance that are similar, and often superior, to more traditional continuous moderate-intensity

exercise (3,5,6,9,18). The exact mechanisms promoting these seemingly potent adaptations to HIIT are not yet fully known but may be related to the training time spent at or near $\dot{V}O_{2max}$, a high degree of muscle fiber recruitment, and other related cardiovascular and cellular signaling pathways. Table 1 provides an overview of the benefits of HIIT, and the next sections highlight some key findings demonstrating that HIIT can promote physiological adaptations that are linked to improved health.

Aerobic Fitness

Because of the well-known benefits of vigorous exercise for improving cardiorespiratory fitness, it is not surprising that aerobic HIIT leads to improvements in $\dot{V}O_{2max}$ and various indicators of the oxidative capacity of skeletal muscle. However, studies in clinical populations have shown that higher-volume HIIT programs using longer intervals lead to greater improvements in $\dot{V}O_{2max}$ when compared with continuous moderate-intensity training of equal volume and time commitment (5,18). These studies also have shown superior improvements in markers of muscle mitochondrial content after HIIT and potent effects of HIIT for improving $\dot{V}O_{2max}$ after as little as 2 weeks of training. Data with lower-volume HIIT protocols are less consistent, with 2-week programs failing to induce aerobic fitness changes that are observed more readily in 6-week programs (3). Together, these findings indicate that some minimum volume or duration of training may be a key feature in the benefits related to HIIT.

Metabolic Health

Improvements in insulin sensitivity and glucose control after aerobic exercise play a key role in the prevention and management of type 2

TABLE 1: Summary of Physiological Benefits of HIIT

Aerobic fitness	HIIT increases $\dot{V}O_{2max}$ to a greater extent and more quickly than continuous moderate exercise.
Metabolic health	HIIT improves insulin sensitivity more effectively than continuous moderate exercise, especially in individuals at risk for developing diabetes mellitus.
Cardiovascular health	HIIT improves endothelial function more effectively than all forms of continuous exercise.

References for above: 5,8,11,19.

DM and related conditions (e.g., metabolic syndrome, prediabetes). Because replenishment of muscle glycogen is a key factor mediating the improvement in insulin sensitivity after acute exercise (11), it is hypothesized that greater depletion of muscle glycogen after aerobic HIIT may facilitate enhanced muscle glucose uptake and improved insulin sensitivity. Importantly, research indicates that HIIT can increase insulin sensitivity and improve glycemic regulation, particularly in those with or at risk for type 2 DM. Two weeks of low-volume HIIT produced significant improvements in insulin sensitivity in overweight individuals (11). Less extreme HIIT protocols, which likely are more practical for individuals with metabolic diseases to perform, also have shown promise for improving insulin sensitivity along with glucose control. Research indicates that 2 weeks of HIIT training using the practical model produce improvements in insulin sensitivity in type 2 DM patients (3,9). Other HIIT protocols also seem to improve glycemic control and may be superior to continuous moderate-intensity exercise. One recent 12-week training study indicated that high-volume interval walking was more beneficial for glucose control than continuous moderate exercise despite no differences in total energy expenditure (8). This study also reported greater benefits of interval walk training for reduced abdominal visceral fat mass, a key component of metabolic health.

Vascular Health

The benefits of aerobic exercise for improving the health and function of blood vessels is linked to endothelial cells within the vasculature, which release nitric oxide and have potent vasodilating effects to protect cardiovascular function and health. The proper function of these endothelial cells ensures efficient blood flow distribution. Research studies clearly demonstrate greater improvements in endothelial function after HIIT when compared with continuous exercise. These findings appear consistent across a range of cardiovascular disease states and also have shown that HIIT can improve the elasticity of arterial blood vessels (5,18).

PSYCHOLOGICAL RESPONSES

Although much research has examined the physical health benefits of aerobic HIIT, very little is known about how people



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perceive HIIT, whether they like it or dislike it, whether they can tolerate it, and most importantly, whether they will adhere to HIIT. Knowing how people respond perceptually and behaviorally to HIIT is of paramount importance if this approach to exercise is to be encouraged for use by clients and patients. The next sections consider what currently is known about the impact of aerobic HIIT training on affect, enjoyment, and adherence.

Affective Responses

“Affect” is a response that is elicited somewhat reflexively or instinctively without significant thought and is linked to pleasure or displeasure and tension or calmness (12). Clearly, aerobic HIIT is not a viable approach to exercise if clients and patients perceive it as aversive (*i.e.*, painful, displeasurable, stress provoking), which is the common outcome of prolonged exercise at high intensities. Although popularized media programs give the impression that HIIT is entirely grueling, interval exercise need not be a displeasurable experience. A built-in benefit of HIIT with respect to affect is the recovery period, which provides exposure to lower-intensity exercise, which is known to produce more favorable responses. So what about affect experienced during the work intervals, where people are exercising at very high intensities? Research indicates that affect experienced during these intense bursts is not different from the affect experienced during continuous moderate-intensity exercise and is significantly more positive than continuous vigorous-intensity exercise (7,14). These responses tend to be consistent across a relatively wide range of demographics including healthy young adults and middle-aged adults with prediabetes or DM. Furthermore, this research suggests that the “most pleasurable interval” might be relatively short (60 seconds or less) and near-maximal (approximately 90% of aerobic capacity).

Enjoyment Responses

Whereas affect is an instinctive mood response, enjoyment is a psychological state that is unique in that it is elicited after appraising or cognitively evaluating a situation. Enjoyment considers the answers to questions like: Do I like that feeling? Do I feel accomplished? Do I feel revived or refreshed? Or more globally, did I enjoy my exercise experience? This focus on enjoyment is relevant because of concerns that vigorous-intensity exercise may be less enjoyable and, thus, less well tolerated than moderate-intensity exercise. Although this concern related to intense exercise is true for continuous exercise (*e.g.*, walking for 30 minutes typically is perceived as more enjoyable than fast running for 30 minutes), recent data suggest that vigorous-intensity exercise performed in an intermittent fashion is very enjoyable (2,7,14). When comparing aerobic HIIT with continuous moderate-intensity exercise and continuous vigorous-intensity exercise, participants report highest enjoyment for HIIT conditions (2,7,14).

Adherence

Although research indicates positive affective responses and perceived enjoyment of HIIT, questions related to long-term



exercise adherence remain. There is, however, a limited amount of data demonstrating that affective responses during exercise are better able to predict exercise adherence than the responses that occur after the completion of exercise (17). This research makes clear that how we feel during exercise is important. Aerobic HIIT does appear promising in that it has been viewed as equally or more favorable during the exercise bout than continuous moderate-intensity or continuous vigorous-intensity exercise, although these results are preliminary.

SAFETY CONSIDERATIONS

Although HIIT may be both potent for health and generally well tolerated, participation in this form of exercise requires practitioners to consider established guidelines related to risk (1). Specifically, this form of training fits somewhat loosely within the umbrella of vigorous exercise and therefore is only appropriate for low-risk individuals, moderate-risk individuals who have been cleared for vigorous intensities by a medical professional, and high-risk individuals who are under direct medical supervision during exercise training. A related risk consideration is linked to exercise modality. Most of the research to date in this area has been conducted on cycle ergometers. Importantly, exercising on a treadmill poses a greater risk of injury via falling than cycle-based exercise. Practitioners should, therefore, remind clients and patients to be cautious while performing aerobic HIIT. One specific recommendation might include inducing high intensities by adjusting treadmill grade rather than speed, which can pose additional risk for those unaccustomed to fast running on a treadmill (see Sidebar).

SIDEBAR: TIME EFFICIENCY

Some of the excitement surrounding HIIT exercise is that it can deliver powerful outcomes with relatively low volumes of exercise and a minimal time commitment. Although there is much potential for this type of claim, it is important to consider that each form of HIIT is different in the amount of time spent doing heavy exercise and the total amount of time spent completing the session.

TABLE 2: Recommendations for Practitioners

Educate	Provide clients and patients with information related to the benefits of HIIT.
Base prescription	Warm-up at a low to moderate intensity for 2 to 5 minutes. First complete a 1-minute high-intensity work interval at about 90% of max effort. Then complete a 1-minute low-intensity recovery interval at about 10% max effort. Repeat the work and recovery intervals 10 times for a total of 20 minutes. Cooldown at a low to moderate intensity for 2 to 5 minutes.
Start slowly	Increase the number of HIIT sessions into an exercise program across time.
Trial and error	Encourage clients to try a variety of HIIT workouts to find one that is enjoyed. Consider manipulating the duration of the intervals to shorter or longer periods. Adjust the recovery period for more or less rest between work intervals.
Change it up	Try HIIT exercises indoors and outdoors, with music and without music, alone and with workout partners, etc.
Be careful	Reinforce that intense exercise increases risk and to be cautious when trying new HIIT routines, especially those that require any unorthodox movement patterns.

HIIT RECOMMENDATIONS

The growing amount of research related to aerobic HIIT should provide practitioners with increased confidence in prescribing this form of exercise, especially versions that focus significantly on aerobic-type activities. Furthermore, utilization of aerobic HIIT seems both feasible and valuable for a range of individuals — from diabetic patients to devoted fitness enthusiasts and every group in between. The task of the health and fitness practitioner is to recommend HIIT as an option for the individuals they serve in their daily work. Part of this process is the recognition that HIIT comes in many forms and that a “one size fits all” approach likely is not possible. Table 2 provides several recommendations and a sample prescription for practitioners on how they might go about encouraging HIIT exercise for their clients and patients.

A final note on how HIIT exercise can fit into a long-term plan for health and wellness links to motivation. As many health and fitness practitioners know, motivation derived from the desire to achieve a lofty outcome goal is perhaps less facilitative for long-term adherence than goals and motivations linked to the process of the behavior. Specifically, outcome goals such as weight loss or improving $\dot{V}O_{2\max}$ represent a focus on what exercise can deliver across time. In contrast, process goals such as having a pleasurable experience and sensations of accomplishment from exercise sessions may produce more desirable long-term adherence as they encourage the exerciser to take note of their positive affective experience. Therefore, practitioners are encouraged to focus client and patient attention on these near-term beneficial aspects of aerobic HIIT-based exercise.

SUMMARY

Experienced health and fitness practitioners know that fitness fads come and go quickly. However, it is unlikely that the current fascination with interval training will fade away given

the growing scientific evidence in support of HIIT. The many potential positive benefits described within this article make clear that aerobic HIIT programs provide great benefit to health. However, more research is needed to assess body weight/resistance versions of HIIT. Furthermore, there is work to be done to examine the impact of HIIT on acute psychological responses and exercise adherence. For now, however, practitioners are encouraged to consider HIIT as a valuable tool in the promotion of health and fitness in a wide variety of populations.

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Disclosure: The authors declare no conflicts of interest and do not have any financial disclosures.



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BRIDGING THE GAP

Aerobic HIIT is an effective and proven approach to fitness training that is known to benefit aerobic fitness, metabolic health, and cardiovascular function. Such training requires the exerciser to engage in exercise at intensities that are known to have negative impacts on mood. However, emerging research suggests that this form of training is pleasurable and may provide a viable training option for individuals desiring to improve health and fitness.